



Bangladesh University of Business and Technology

Course Title: Computer Networks Lab

Course Code: CSE 320

LAB REPORT

Experiment Name: Static Routing

Experiment No.: 04

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Introduction

Static Routing is a networking method where paths are defined by the network administrator and remain fixed until manually updated. In static routing, traffic paths are predictable and strictly controlled by the admin, offering high security, low overhead, and stability. This makes static routing ideal for small and simple networks.

In this experiment, we will implement static routing using a set of network devices (routers and switches) and client hardware.

Devices and Tools

- Cisco Packet Tracer v9.0.0
- 2960-24TT Switch × 6
- ISR4331 Router × 3
- Client PC × 12

Client Addressing Table

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC3	192.168.2.11	255.255.255.0	
PC4	192.168.3.10	255.255.255.0	192.168.3.1
PC5	192.168.3.11	255.255.255.0	
PC6	192.168.4.10	255.255.255.0	192.168.4.1
PC7	192.168.4.11	255.255.255.0	
PC8	192.168.6.10	255.255.255.0	192.168.6.1
PC9	192.168.6.11	255.255.255.0	
PC10	192.168.7.10	255.255.255.0	192.168.7.1
PC11	192.168.7.11	255.255.255.0	

Router Interfacing Table

Device	GigabitEthernet0/0/0	GigabitEthernet0/0/1	Serial0/1/0	Serial0/1/1
Router0	192.168.1.1	192.168.2.1	192.168.5.1	N/A
Router1	192.168.3.1	192.168.4.1	192.168.5.1	192.168.8.1
Router2	192.168.6.1	192.168.7.1	192.168.5.3	192.168.8.2

Procedure

The following steps were meticulously followed to setup static routing among the network and client devices:

Preparing the Network:

- In the Cisco Packet Tracer application, we add three new **Routers (ISR4331)** and two child **Switches (2960-24TT)** for each of them. Then we add twelve Client PCs (PC0-PC11). We connect two clients with each of the switches. We connect the network devices using the **Auto Cabling** tool.
- We also interconnect **Router0, Router1, and Router2** using the **Auto Cabling** tool.

Configuring the client PCs:

- Single click the **PC0** and a new window appears.
- Under the **Desktop** tab, go to IP Configuration.
- Add the **IPv4 Address, Subnet Mask, and Default Gateway** according to the aforementioned **Client Addressing Table**. Repeat the steps for **PC1-PC11**.

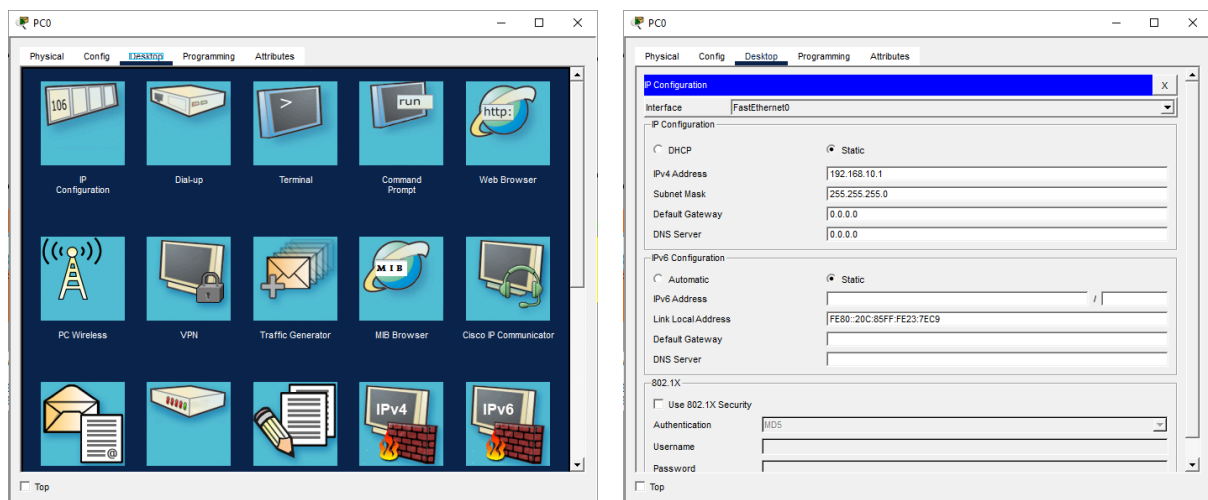


Fig 1.1: Configuring the client PCs.

Configuring the Routers:

- Click **Router0** and a new window will appear.
- Under the **Physical** tab, we turn off the router. Then we select the **NIM-2T** module from the left sidebar and place it onto the router's physical view. We turn on the router and give it a few minutes to boot.

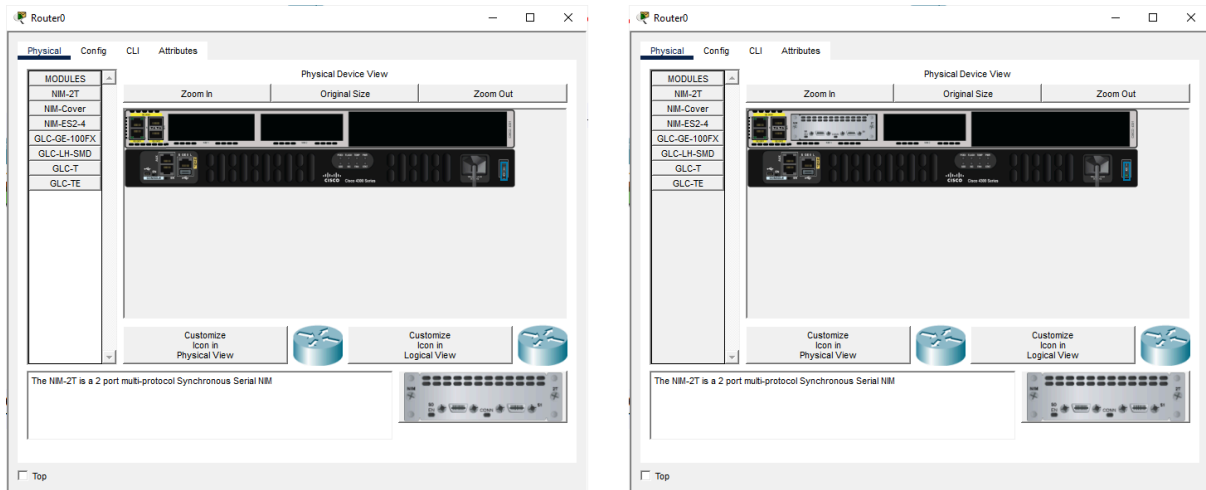


Fig 1.2: Configuring the Physical Modules.

- Under the **Config** tab, we configure the IP addresses and Subnet masks of **GigabitEthernet0/0/0**, **GigabitEthernet0/0/1**, **Serial0/1/0**, and **Serial0/1/1** according to the aforementioned **Interfacing Table**. We make sure to turn on the port.

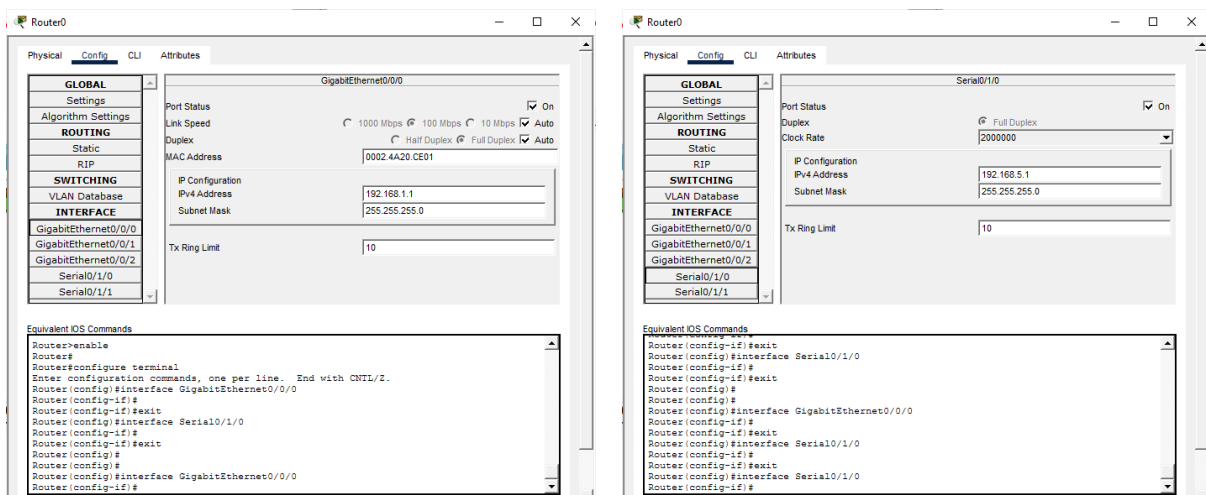


Fig 1.3: Configuring the Interfaces.

- We navigate to the **Static** section and configure the routes. Static routes define the traffic via hop addresses.

For example: the Router0 has two subnets: 192.168.1.1-2 and its neighbouring Router1 also has two subnets: 192.168.1.3-4. Finally, Router2 also comes with two subnets: 192.168.1.6-7. Each of these addresses are considered traffic routes and have to be added manually.

In the static routes fields, we add “192.168.1.1” as Network, “24” as Mask, and Next Hop to 192.168.5.2 (Router1’s IP address) to establish a single connection. Then we repeat the process for each route logically possible in our network setup.

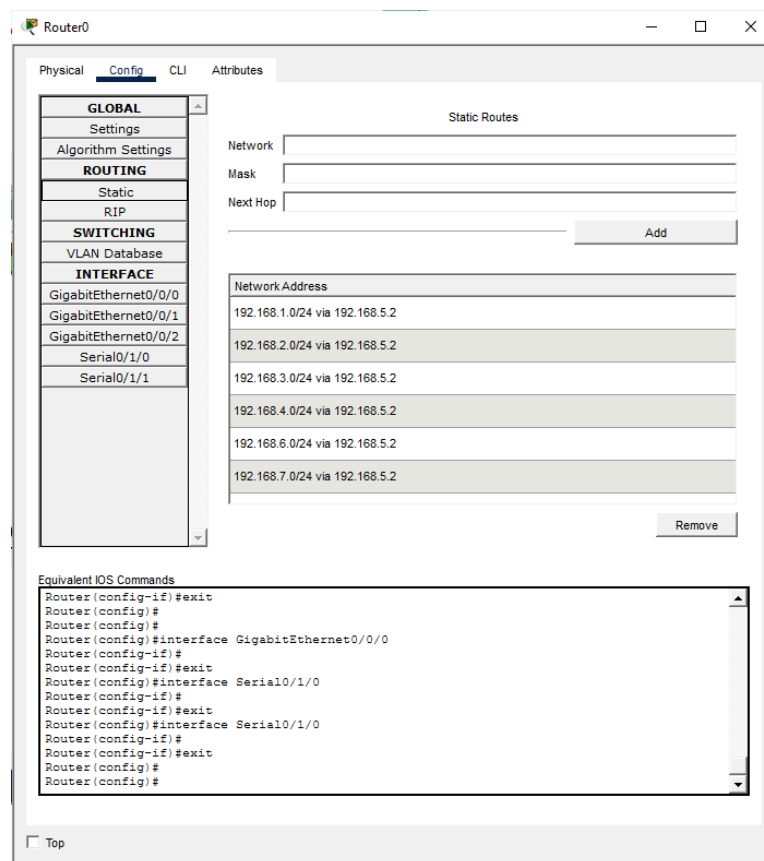


Fig 1.4: Configuring the Static Routes.

- We repeat all the steps for Router1 and Router2.

Result

Once we have followed all the steps mentioned in the previous sections, we should have a working static routing configuration of three Routers, six Switches and twelve client PCs:

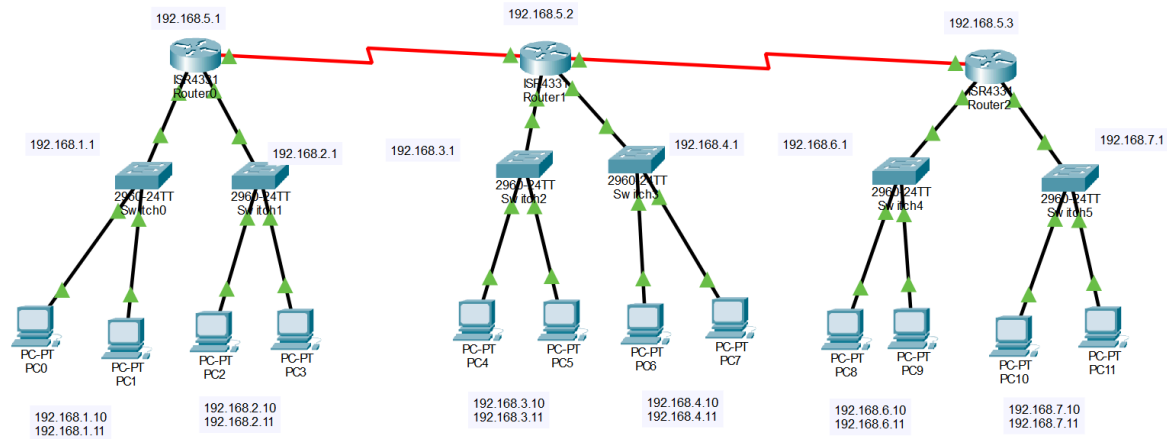


Fig 2.1: Static Routing.

Testing and Verification

We send messages from client to client to check whether the data is being transmitted properly, ensuring that **Static Routing** is properly implemented.

- From the toolbar, click “Add Simple PDU”. The cursor would turn into a message icon.
- Click on PC0 and then under the same router, click PC2.
- If everything goes as expected, we should see a successful communication.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
...	Successful	PC0	PC2	ICMP	...	0.000	N	0	(e...	
...	Successful	PC1	PC3	ICMP	...	0.000	N	1	(e...	
...	Successful	PC4	PC5	ICMP	...	0.000	N	2	(e...	
...	Successful	PC8	PC10	ICMP	...	0.000	N	3	(e...	
...	Successful	PC0	PC2	ICMP	...	0.000	N	4	(e...	

Fig 3.1: Testing communication via static routing.

Conclusion

The objective of the fourth experiment was successfully achieved by configuring three different Routers, six Switches, and twelve client Devices to implement a network with static routing. It provided core skills for implementing static routing for small scale, simple networks.
